



Towards optimal positioning and energy-efficient UAV path scheduling in IoT applications

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ABSTRACT

The Unmanned Aerial Vehicle (UAV) based communication has been emerged as a feasible solution for remote applications such as disaster management, and search and rescue due to its mobility and cost efficiency. The prior researches in this domain focused on positioning and path planning of UAVs; however, these approaches faced several limitations due to adverse weather conditions of the environment. In this paper, the impact of weather-based positioning and path planning of UAVs (IWPOP-UAV) is carried out to achieve increased QoS, reliability and energy efficiency in UAV communications. Initially, the prediction of weather conditions in the emergency situations is performed by utilizing the Cerebral Long Short-Term Memory (C-LSTM) which possesses negligible training loss and increased accuracy. The cell-based partitioning of emergency area is carried out in order to determine the target UEs. The decision upon number and position of the UAVs in each cell is provided by the A3C algorithm based on weather conditions and other significant factors thereby achieving increased coverage ratio and minimal requirement of UAV transmit power. The path planning of the UAV in order to perform effective collection of data is considered as a multi-objective optimization problem and executed by using Mayfly Optimization Algorithm (MOA). By doing so, the proposed approach is able to achieve increased QoS, reliability and energy efficiency in UAV based communication. The proposed IWPOP-UAV approach is experimented in NS 3.26 and evaluated in terms of performance metrics such as coverage ratio, cell coverage, delay, path gain, number of collected packets, UAV transmit power, and energy consumption. The results obtained are summarized and concluded to demonstrate the efficacy of the proposed approach.

1. Introduction

The Unmanned Aerial Vehicle (UAV) has been emerged as a significant technology in many fields of works such as disaster management, geographical mapping, delivery services, aerial photography, search and rescue etc. [1]. The UAV based base station (UAV-BS) is utilized during the emergency situations in which the ground communication through ground base station is affected by disasters [2]. The deployment of multiple-UAVs has been found to be effective rather than single UAV system [3]. The communication between UAV to ground (A2G) and UAV to UAV (A2A) has been developed to facilitate the applications [4]. The deployment and communication of UAVs is affected

by various factors such as weather conditions like temperature, wind speed, wind direction and UAV altitude etc. [5]. The UAVs work from the limited energy provided by batteries hence the energy consumption should be minimized to lowest possible level; this can be done through proper placement and path planning of UAV [6]. Several existing works considered the effect of wind in positioning and path planning of UAV in order to provide energy efficient and collision free path. However, planning of path only based on distance and disturbance caused due to wind was not effective [7,8].

The more practical experimentation of path planning was achieved by using different wind loads and the optimal control of UAV was

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obtained against all situations. This approach was found to withstand the disturbance caused by wind. Nevertheless; this approach was not energy efficient due to the lack of consideration of several significant environmental factors [9]. The impact of several weather conditions on A2G communication was analyzed and the relationship between channel fluctuation and power received was studied. However, the channel fluctuation caused by UAV altitude was not considered which affects the performance of this system [10,11]. The throughput in A2G communication was improved by optimally positioning the UAV based on the coverage access probability, this allows the UAV to support the maximum number of users in the ground surface. However, the assumption of good weather condition limits the usage of this approach in adverse situations [12,13]. The QoS and QoE of UAV communication were improved by reducing the path loss and time involved in controlling the motion of the UAV. But these methods face scalability issues when number of UAVs was increased [14,15]. The link disconnection problem in indoor environment was overcome by placing the UAV to minimize outage probability. However, the impact of weather conditions on UAV communication was not considered which contributed to disconnection of links [16].

The coverage probability was concentrated to optimally position the UAV in the network in order to obtain efficient A2G communication. Nevertheless, the scalability of UAV causes collision between the UAVs causing damage to UAVs [17]. The path planning of UAV was achieved by implementing various optimization techniques but the inefficiency of these approaches caused convergence in local minima thereby affecting the accuracy of path planning [18]. The efficient path planning was achieved through implementing multi-objective techniques but all these approaches performed any one operation resulting in lack of overall performance of controlled UAV system [19]. The channel modeling of A2G and A2A communication was performed to provide optimal data collection thereby improving the connectivity in the network [20,21]. The positioning of UAV and path planning was carried out by various approaches however there is still a demand for an effective approach to perform joint optimization of positioning and path planning by considering the weather conditions in order to achieve optimal QoS in both normal and emergency situations. The major research questions in this area are formulated as,

- RQ1: How to achieve QoS and reliability in UAV communication?
- RQ2: How to achieve increased lifetime of UAV communication during emergency situations?

1.1. Research motivation

The major aim of this research work is to provide improved connectivity to the UAV communication during adverse situations by analyzing the impact of weather conditions in various operations of UAV communication. The overall control and energy efficiency of the UAV system is enhanced by performing proper deployment and path planning of UAV. UAV communication encounters lots of challenges due to its dynamic topology and three dimensional characteristics. Several existing works provides individual solutions to the challenges but these solutions are not effective and cannot be adopted in all situations. The important research problems are listed below,

- Dynamic weather conditions

- (1) Wind effects — the placement and communication of UAV is significantly affected by various wind characteristics such as wind direction, wind speed, wind velocity, wind shield, propeller vortex and turbulent flow. All these characteristics have specific impacts on the operation and communication channel of UAVs.

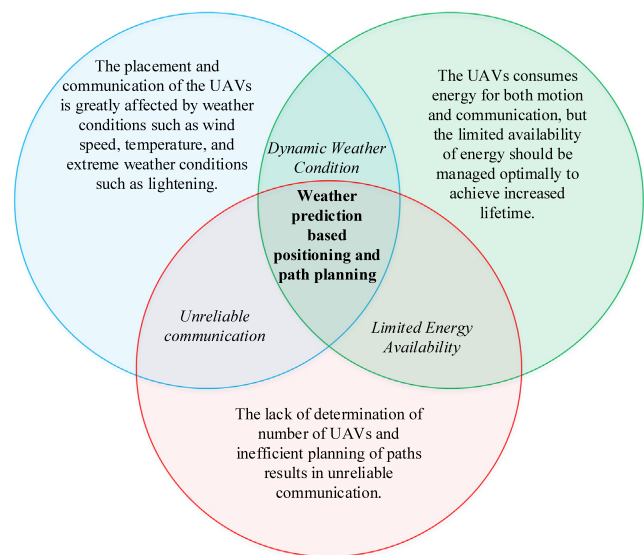


Fig. 1. Diagram of research motivation.

- (2) Temperature — Both the extreme cold and extreme hot temperatures affects the operation of UAVs. The cold temperature affects the UAV batteries and also when the moisture enters the UAV it could even cause short circuiting leading to crashing.
- (3) Extreme weather conditions — The extreme weather conditions such as rain, fog, snowfall, hail and lightening affects the UAV communication and can even cause physical damage to the UAV.

- Limited energy availability — the UAVs operate with the energy provided by the batteries attached to it. This energy is used for both UAV movement and communication. Hence the energy consumption is to be minimized to avoid frequent charging issues.
- Unreliable communication — The UAV performs collection of data from the UEs, but the lack of determination of number of UAVs required for reliable communication affects the data collection process which results in reduced QoS and unreliable communication.

These issues are considered in order to formulate the objectives of this approach which are also shown in Fig. 1. The major objective of this research work is to deliberate the effects of weather conditions in UAV communication during adverse situations. Both A2G and A2A communication is performed to achieve improved QoS and enhance the connectivity in the environment. This objective is achieved by accomplishing the following sub objectives which are listed as,

- To analyze the impact of weather condition by accurately predicting the current weather information from past historical data.
- To maximize the coverage area of the UAV by deploying the UAV considering several environmental factors.
- To maximize the energy efficiency of the UAV and to minimize the time consumption by optimally planning the trajectory.

1.2. Research contributions

This research work concentrates on positioning and path planning of UAVs based on the predicted weather conditions in adverse situations. The major contributions of this research approach are mentioned as follows,

- The impact of weather condition on effective UAV communication is analyzed and the prediction of weather conditions such as temperature and wind characteristics is taken place by implementing Cerebral LSTM.
- The environment is constructed as hexagonal cells and the cell partitioning is executed based on number of users and distance between the users and the entropy function is used to determine the number of partitions for effective communication.
- The positioning of multiple UAVs is carried out by the Asynchronous Actor Critic (A3C) algorithm in which the number of UAVs and their respective positions is determined for each partition.
- The energy efficient path planning is implemented by using Mayfly optimization algorithm in which the optimal planning of path is achieved based on several parameters thereby increasing the lifetime of the UAV. In the validation process, the performance of the proposed approach is evaluated in terms of several performance metrics in order to prove the efficacy of our approach.

1.3. Paper organization

The remaining of the paper is organized as follows; Section 2 provides the preliminary knowledge about the application of UAV in 5G communication. Section 3 provides the literature survey in deployment and positioning of UAVs along with the path planning of UAVs. Section 4 describes the problem statement of the positioning and path planning of UAVs. Section 5 demonstrates the research methodologies involved in the proposed approach with necessary explanations, architectures, and mathematical expressions. Section 6 performs the experimentation of the proposed approach in which the proposed approach is validated in terms of several performance metrics in order to prove its significance. Section 7 deals with concluding the significance of proposed approach along with the future work directions.

2. Preliminaries

This section explains the preliminary knowledge of this research. It explains the overview of the 5G enabled UAV environments; in addition, it demonstrates the communication between the UAV and base station.

2.1. 5G enabled UAV environment

The emerging field of UAV has many significant processes within the wireless networks. Due to the developments in cellular technologies, UAV integrated with 5G network with the aim of enhancing the communication of UAV. Low latency and reliable communication is the important characteristics in UAV communication for efficiently control the UAV. This integration system provides low latency and high throughput services. The cellular connected UAV are estimated to controller and managed by Ground base station (GBS). The 5G enabled UAV network provides three types of services such as Ultra reliable low latency communication (URLLC), extreme mobile broadband (eMBB), and massive machine type communication (mMTC) [22]. The URLLC service is most important scenario in 5G enabled UAV. Due to the modification of distance between UAV and GBS, velocity, altitude, the URLLC service may be vary. Cellular connected UAV has the advantage of high data transmission form UAV to GBS. In heterogeneous environment the URLLC application satisfies the both requirements such as latency and reliability [23].

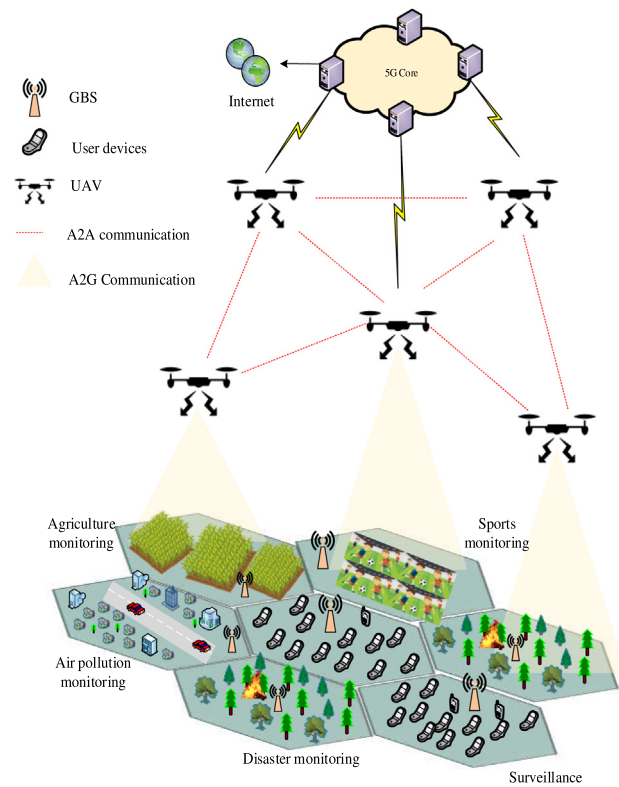


Fig. 2. 5G enabled UAV environments.

2.2. UAV with cellular communication

Generally, UAV has flying base stations which can intellectually reposition by itself for assisting the previous terrestrial communication system that helps to improve the Quality of Experience (QoE) and coverage. The base station mounted on the UAV can provide the demand that provides the solution for emergency situation and disaster management. If the ground base stations (GBS) are affected by disasters then the UAV uses its flying base stations. This type of UAV to GBS communication is known as air to ground communication (A2G). 5G base stations improve the coverage area that also increases the data demands in 5G cellular systems.

2.3. UAV-UAV communication

In this communication system the UAVs communicate directly with others for sharing the spectrum of cellular with users for autonomous flight behaviors, and collision avoidance. UAV to UAV communication influences similar frequency spectrum with uplink of ground users. This communication is known as air to air communication (A2A).

Fig. 2 represents the 5G enabled UAV environments, in which the UAV monitors the environment and collects the information using cellular communication. Table 1 represents the applications and roles of UAV.

In A2A communication the UAV shares the spectrum of cellular with users. A2G communication collects the environment information for take action in a particular time. 5G core network provides low latency and reliable communication to the UAV.

3. Literature survey

This section explains the literature survey based on UAV positioning and path planning in 5G-enabled UAV environment. In addition, research gaps are analyzed in a brief manner.

Table 1
Applications of UAV.

Areas	Examples	Role of UAV
Military	Searchers for injured or lost soldiers	Live streaming communication for sending videos to ground groups
Environmental applications	Climate change	Information collection using sensors
Agriculture	Crop monitoring and soil analysis	Information captured by the drones in agriculture field
Public safety communication	Voice communication in disaster situations	Act like aerial base stations
Photography	Captures the events like, sports, competitions, social gatherings	Capture videos using cameras
Law enforcement surveillance	Search and rescue	Capture the videos using cameras for surveillance

3.1. Schemes for UAV deployment and positioning

Authors in [24] proposed a method for positioning of UAV by implementing deep learning approaches to achieve maximization of throughput. The estimation of throughput was carried out based on data traffic and UAV capacity. The access coverage probability of air-to-ground channel link was computed by considering the RSSI, SNR, distance, elevation angle and altitude. The K-means algorithm was used to cluster and classify the input parameters into three different classes. The hybrid Multi-Layer Perceptron and Long Short Term Memory (MLP-LSTM) was proposed to predict the positioning of the UAV based on the inputs. The research gaps of this research are discussed as follows, the positioning of UAV was determined to achieve maximum throughput but in multi-UAV scenarios other factors like collision avoidance should also be concentrated. The weather condition was assumed to be good but in real time scenarios extreme weather conditions also exists which will affect the channel limiting the adoption of this approach.

Authors in [25] proposed an optimization strategy for the deployment of UAV base stations in 3D to achieve improved QoS. The necessity for deployment of UAV base station was addressed and the number of covered users was increased in order to achieve the QoS requirements. The QoS was measured by keeping in view of allowed path loss and requirement of particular range of data rate. The Genetic Algorithm (GA) was implemented to solve the optimization of 2D deployment of UAV to achieve better QoS. The limitations of this research are defined as follows; the altitude determination was performed in an inefficient way, results in variation in optimal altitude of the UAV which leads to coverage area problems. The genetic algorithm was implemented to deploy the UAV which is complex optimization algorithm with increased consumption of time which is a drawback in emergency situations. An optimized deployment strategy of drone base station is proposed in order to improve the user experience [26]. The optimal location of drones was determined by computing the moving speed of drones from one position to another, service time, control time for the drones. The antenna array of the ground users was considered to position the drones. Based on the estimated angles the position of base station was determined by implementing the greedy algorithm. Limitations of this research are explained as follows, only initial deployment of drone position was carried out but when the number of users in the ground varies this approach faces scalability issues. The implementation of conventional greedy algorithm made this approach inefficient as this algorithm has major limitation of failing to obtain the optimal solution in all iterations.

Authors in [27] proposed an approach for efficient communication and control of wireless drone based antenna array. The stabilization of drone movement was achieved by reducing the service time and control time associated with the movement of drone. The proposed positioning approach was experimented in simulation tool with 100 users and 10 drones and evaluated to prove that this method outperforms other existing approaches in terms of reduced control time and improved stability.

However, the number of users is dynamic in nature and positioning of drone base station only based on number of users is not efficient as it affects the spectral efficiency. The presence of blockage is also not considered which affects the accuracy of the approach. Resource allocation and location decision strategies are proposed for UAV to obtain reliable connection in emergency indoor communication [28]. The necessity of reliable link for indoor users during emergency situations was addressed and the deployment of UAV for that purpose was carried out. The link disconnection in indoor environment was caused due to signal fading and increased outage probability. The optimal positioning, power and bandwidth allocation was achieved by implementing LBP-fairness algorithm. However, the positioning of the UAV base station was presented to provide connectivity to indoor users but the effect of adverse weather conditions on the channel link between UAV and user also affects the connectivity.

A game theoretic approach is proposed for the deployment of multiple UAVs to obtain optimal coverage [29]. The directional antennas of UAVs were used to provide coverage for multiple users in the ground surface. The payoff value in the game theoretic approach is mainly based on coverage probability of the UAV. The altitude of UAV was altered to obtain optimal coverage probability in order to increase the connectivity. However, the coverage probability was concentrated to improve the coverage area in then network but in multi-UAV scenarios the positioning of UAV only based on coverage area causes collision between UAVs. Deep learning based fifth generation millimetre wave communication channel tracking is proposed for unmanned aerial vehicle internet of things networks [30]. The channel tracking between UAV and the user node was facilitated by using sensors and antennas. The deep learning based deep auto encoder (DAE) was used to estimate the altitude of the UAV in order to produce efficient beamforming. The altitude estimation was based on three inputs namely accelerometer, gyroscope and magnetometer which were placed in the IMU of the UAV. The altitude estimation was provided as angle measurements of roll, yaw and pitch angles respectively.

3.2. Schemes for UAV path planning

Authors in [31] proposed an effective path planning approach for UAV based on improved particle swarm optimization. The path planning of UAV from source to destination is provided in order to increase the connectivity and reduce the time and energy consumption involved in it. The collision in the multi-UAV situation is eliminated by formulating the objective functions such as minimal flying path length cost, radar threat cost, mountain terrain cost and UAV collision cost. An effective mutation strategy was also adopted in order to increase the convergence speed. However, the path planning was carried out using improved Particle Swarm Optimization (PSO) for better convergence speed and collision avoidance but the presence of obstacle in the environment was not considered resulting in inefficient connectivity.

A novel hybrid grey wolf optimizer algorithm is proposed for path planning of UAVs [32]. The necessity of optimal path planning of

Table 2
A summary of related works.

Approaches	Algorithm/ Process	Research gaps
UAV Deployment and Positioning	MLP-LSTM [24], GA [25], Greedy [26], Bang theory [27], LBP [28], Game theory [29], DAE [30]	• Achieve less throughput due to lack of collision avoidance • Variation in optimal altitude of UAV leads to coverage problem • Initial deployment of drone position leads to scalability issues • UAV positioning is only based on coverage area that leads to collision between UAVs. • Implementation of conventional algorithms made this process inefficient and also it does not obtain optimal solution in UAV positioning
UAV Path Planning	IPSO [31] SWG0 [32] QoS-Path [33] DQN [34] MSA [35] Priority-data [36] ANN [37] RCS [38]	• Inefficient connectivity due to presence of obstacle in the environment • Obtain less QoS due to lack of the weather conditions. • Poor path planning due to lack of weather condition, presence of blockage and limited energy availability

UAV is insisted and an efficient hybrid algorithm comprising of Simplified Grey Wolf Optimizer (SGWO) and Modified Symbiotic Organisms Search was proposed. The GWO was combined with SOS in order not to be stuck in the local minima and to accelerate the speed of convergence. The exploitation phase was improved by modifying the commensalism phase in an effective manner. However, the hybridization of two optimization algorithm for effective path planning increases the complexity and time consumption thereby increasing the energy consumption of UAVs. Authors in [33] proposed a QoS aware coverage path planning approach in 5G mmWave network. The major problems affecting the transmission of data and coverage of camera from UAV to ground user was found to be presence of blockage, UAV movement and variation in channel parameters. The path planning of UAV to obtain the improved QoS in network was formulated as a nonlinear optimization problem further this problem is divided into several sub problems and solved individually. The proposed path planning approach was experimented as full stack simulation and the parameters such as end-to-end throughput and path loss was evaluated. However, the effective path planning was carried out to obtain improved QoS but the channel between UAV and user also gets affected due to weather conditions which also affect the QoS of the network.

Distributed energy efficient multi-UAV navigation system is proposed for long-term communication coverage by deep reinforcement learning [34]. The usage of UAV base station in emergency situations where the regular infrastructure of the environment is collapsed was addressed and the coverage of the UAV is improved in an effective way to achieve long term communication. The navigation of UAV was formulated as Partially Observable Markov Decision Process and was solved using Deep Q Network (DQN). Multi-objective UAV path planning approach is proposed for emergency information collection and transmission [35]. The need for mobile base station during the emergency situation was addressed and an effective way point method was adopted. Initially, the users in the ground surface are clustered using mean shift algorithm (MSA) and the positioning of UAV in 3D was carried out based on the user location. Then the channel modeling for wireless transmission is carried out by performing LoS and NLoS estimation. Various constraints such as wind effect and terrestrial constraints were considered and the parameters such as cell coverage, channel capacity and path gain were improved by implementing waypoint based path planning.

Authors in [36] proposed a priority based data collection for UAV aided mobile sensor network. The path planning of the UAV was based on the process of data collection in which the two mechanisms namely Prioritized Based Contact Duration Frame Selection Mechanism (PCDFS) and prioritized Based Multiple Contact Duration Frame Selection Mechanism (PMCDIFS) were proposed. The stable motion of UAVs based on the priority user group was facilitated without any collision

during the collection of data. The waypoint model was leveraged to precede easy path planning process. At first every user node joined the UAV with a join message and begun to transmit the packets. The collision free efficient transmission was facilitated by implementing balance algorithm.

An adaptive neural network control and optimal planning of UAV with energy consumption prediction is proposed in this research [37]. The necessity for energy efficient path planning of surveillance UAV was addressed. Here, initially the control of the UAV kinematics and propulsion was carried out by using Adaptive Neural Network (ANN) which controls the altitude of the UAV system. The surveillance data about the ground surface were clustered by means of K-agglomerative clustering in order to reduce the complexity of monitoring. Further, the optimal path planning of the UAV was executed by implementing the integration of A* and Set based Particle Swarm Optimization (S-PSO). This provides increased disturbance rejection control with less energy consumption. A novel real time penetration path planning for stealth UAV in 3D complex environment is proposed [38]. The real time path planning was provided by integrating the real time learning A-star algorithm with predictive control named as improved A-star algorithm. The kinematics of UAV along with constraints such as heading angle, pitch angle and roll angle were formulated and the dynamic Radar Cross Section (RCS) model was computed. Further, the path planning was executed by utilizing both A-star algorithm and D-star algorithm which was the integration of A-star with Dijkstra algorithm. Table 2 summarizes the related work and its research gaps.

4. Problem statement

The studies performed in previous sections performed either positioning or path planning of UAV but failed to jointly optimize both the factors by considering the weather information to achieve optimal QoS. In our approach, the joint optimization of positioning and path planning of UAV by considering the weather factors is carried out in order to minimize the time consumption and energy consumption of the UAVs. Let the number of UEs in the emergency situation is denoted as $UE_n = \{1, 2, \dots, k\}$, $n \in [1, b]$ types of equipment in the emergency situation. The number of UAVs be denoted as $UAV = \{1, 2, 3, \dots, b\}$ which is utilized as BS during emergency situations. The variable $\delta_{kb} \in \{0, 1\}$ denotes that the user k is under the coverage of UAV b or not. If $\delta_{kb} = 0$ then k is not covered by b, and if $\delta_{kb} = 1$ k is covered by b. The 3D positioning of UAV is divided into two processes in which the 2D position of the UAV is carried out initially after which the altitude of the UAV will be defined. This is due to the fact that the altitude affects only the path loss of the channel. Once the coverage of UAV is determined the altitude is further explored to achieve minimum path loss.

The coverage of the UAV b can be formulated as,

$$\|a_b - \delta_{kb}c_k\| \leq CR_b + V(1 - \delta_{kb}) \quad (1)$$

Where, a_b denotes the horizontal position of the b , c_k denotes the horizontal position of the k . CR_b denotes the coverage radius of b and V denote the huge constant greater than the possible horizontal distance. The allocation of data rate can be formulated as,

$$\sum_{k=1}^{|W|} d_k \delta_{kb} \leq DC_b, b \in \{1, 2, \dots, |U|\} \quad (2)$$

Where, d_k denotes the requirement of data rate by k , DC_b denotes the data capacity of b respectively. The positioning of the UAV can be formulated as,

$$\max_{CR_b, a_b} \sum_{b=1}^{|U|} \sum_{k=1}^{|W|} \delta_{kb} \quad (3)$$

This can be achieved by the certain conditions which represent the optimal coverage of UAV. Eq. (1) represent that if the horizontal distance between k and b is less than CR_b then the user k is under the coverage of UAV b . the second condition for determining the data rate of the UAV b is provided by Eq. (2). The third condition for ensuring that the user k is covered by at most one UAV can be formulated as,

$$\sum_{b=1}^{|U|} \delta_{kb} \leq 1, k \in \{1, 2, \dots, |W|\} \quad (4)$$

The next condition determines that the coverage radii of UAVs are not greater than the predefined maximum coverage radius CR_{max} which can be formulated as,

$$CR_b \leq CR_{max}, b \in \{1, 2, \dots, |U|\} \quad (5)$$

The altitude of the UAV b which affects the path loss (PL) between k and b can be formulated as,

$$AL_b = CR_b \tan(\theta_{max}) \quad (6)$$

The flight time of the UAV is defined as the time involved in motion of the UAV along with the time involved in collection of data from the UE in emergency situation. This can be computed as,

$$F_t = m_t + c_t \quad (7)$$

Where, m_t denotes motion time and c_t denotes data collection time. The energy consumption of the UAV is referred to as the energy consumed for motion of the UAV and energy consumed for collection of data. The total energy consumption can be formulated as,

$$EC_{tot} = m_t \cdot m_p + c_t \cdot c_p \quad (8)$$

Where m_p and c_p are the power involved in motion and transmission of data. The path planning of UAV is carried out to minimize the flight time and energy consumption of the UAV. This can be represented as,

$$\text{Minimize } F_t, EC_{tot} \quad (9)$$

Several existing approaches performed positioning and path planning of the UAVs in order to achieve maximum coverage and minimum energy consumption, however these approaches faced major limitations in achieving the objectives. The positioning and channel modeling of UAVs is performed in [39,40]. In [39] the positioning of UAV was based on traffic loads and uncertain weather conditions and was termed as swarm intelligence method in the deployment of mobile edges (SWIM). This method implements bee swarm intelligence (BSI) in which the UAVs represent the bees. The SWIM comprises of two sections namely traffic modeling and dynamic deployment respectively. In the traffic modeling section, the condition factor was used to model the traffic and estimate the traffic conditions. In the deployment section, the BIS are used for effective deployment and position update purposes thereby resulting in effective transmission of data. The impact of weather conditions and wind fluctuations on the positioning of the UAV was

considered in [40] and based on those different models were proposed in order to reduce the alignment losses. These approaches possess various problems which can be stated as follows,

- The deployment of UAV was said to be based on traffic loads and weather conditions but the impact of these weather conditions in positioning the UAV was not addressed which affects the efficiency of this method.
- The UAVs were deployed to assist effective data transmission to moving vehicles in the ground surface but the inefficient trajectory planning of the UAV results in increased the energy consumption of UAV.
- The BIS algorithm implemented for the purpose of positioning of the UAV has large quantity of fitness computation which increases the time consumption and easily results in convergence in local minima.

In UAV-FSO model the altitude of the UAV was assumed to be static which is not the same in real environment. The fluctuation of the channel will also occur in vertical direction resulting in reduced received power.

The path planning of UAVs is performed in [41,42], and [43] in order to optimize the consumption of energy. In [41] the UAVs were used for delivery of goods to the users without the need of human support. Initially, the customers in the surface area were clustered and the path planning between the nodes was carried out. The effect of wind speed and wind direction on the energy consumption of UAV was analyzed and the mission planning was carried out efficiently to reach out the customer satisfaction. The surface of the environment was constructed as a PRM graph in [42] and the wind characteristics such as density measurements and wind prediction map accuracy were considered. The wind field was designed as a 2D vector field varying with time. The A* algorithm was implemented to determine the path of the UAV based on cost functions such as distance minimization, angle minimization and hybrid function.

The optimal control of hexarotor is achieved in [43] by the proposed path planning model by calculating the signal obtained from state estimation and IMU sensor. The local minima problem is overcome by implementing repulsive potential function. The major problems encountered in these approaches are,

- The mission planning was performed based on limited weather conditions such as wind speed and wind direction but when the temperature factor also affects the energy lifetime of UAV resulting in reduced accuracy.
- The mission planning was performed for only limited number of UAVs and when the number of UAVs increase in order to satisfy the customer needs, the time windowing for pick-up and drop purpose becomes complex and encounters several limitations.
- The planning of path for the UAV is only based on distance and wind disturbance but the other factors such as temperature and altitude also affects the path of the UAV which affects the efficiency of this approach.
- The collision-free trajectory planning was said to be robust in nature because of its feature to withstand in windy environment but in order to plan the optimal trajectory, several significant factors such as weather condition and altitude should be considered which reduces the accuracy of the approach.
- The Lack of consideration of energy consumption and presence of obstacles in the environment path affects the performance of this approach as planning was only based on wind characteristics and path length.

Research solutions: In the following we, discussed the solutions for addressing the problems in this field. The proposed approach considers all these problems and provides solution to them by performing joint optimization of positioning and path planning of UAV based

on the predicted weather conditions. The positioning of multi-UAV is carried out in which the number of UAVs is computed based on user density and data traffic and the positioning of UAV is based on various parameters including weather conditions such as temperature and wind characteristics. Moreover, the impact of weather conditions on positioning, path planning and data collection is addressed in an elaborative manner. The path planning of UAV is carried out by implementing Mayfly optimization algorithm based on several parameters including cell coverage, path gain and channel capacity is improved which minimizes the energy consumption of the UAVs. The impact of variation in the altitude of the UAV is analyzed and the optimal altitude of the UAV is determined which increases coverage without increasing the interference and path loss.

5. System model

In this section, the proposed 5G enabled UAV environment comprises of three entities such as UAV, User Equipment (UE), and 5G ground based station (GBS) which are used for performs various processes for achieving the objective is described in detail. In our proposed work we mainly concentrate on analyzing the impact of weather condition in UAV communication and perform positioning and path planning of Multi-UAV system in order to achieve efficient QoS even during disaster situations. However, disaster regions are affected in data transmission for timely warning of events. The proposed work is subjected for disaster environment where the GBS is overloaded or damaged. Fig. 3 represents the overall system model in which the major processes involved in this paper is highlighted and that are described by follows.

- Weather Prediction
- Density Aware Cell Partitioning
- Multi-UAV Positioning
- Energy Efficient Path Planning

5.1. Weather prediction

The weather condition has significant impact on UAV communication. In particular, the UAV location and communication channel is affected by various weather conditions. Mainly the UAV is disturbed by the wind characteristics such as wind direction, wind speed, wind velocity, wind shield, propeller vortex and turbulent flow. Further the temperature also affects the energy consumption of the UAV. Extreme conditions such as lightening, heavy rain etc. make it impossible to operate the UAV causing physical damage to it. These weather conditions should be known prior in order to plan the communication according to it. The accurate prediction of weather is carried out by implementing C-LSTM which was found to have lower training loss than conventional LSTM network.

The proposed C-LSTM is an improved version of LSTM which has better ability for understanding the time series data when compared to traditional LSTM. C-LSTM used innovative type of recurrent cell to provide the effectiveness of the proposed cell which consist one hidden state $H(t-1)$ and two cell states $C1(t)$ and $C2(t)$. For every time stamp t provides the input $I(t)$ with hidden state $H(t-1)$ and cell states $C1(t-1)$ and $C2(t-1)$ which is known as cerebral LSTM because it has the similarity of human brain cerebral hemispheres. The first unit of C-LSTM is the forget gate $f_g(t)$ and the input gate requires the input to update, the candidate value is stored temporarily before the two cell states values are updated. Every stage $H(t-1)$ and $I(t)$ are received as input and output that is calculated based on the weight values of the parameters. Here, past series data of weather condition (P_{wc}) and current environment data (C_{wd}) are considered as input that is fed into the forget gate of C-LSTM.

Pseudocode for Cerebral LSTM

```

Initialize  $H(t-1)$ ,  $UC(t)$ ,  $LC(t)$ ,  $UT_c(t)$ ,  $LT_c(t)$ ,  $P_{wc}$ ,  $C_{wd}$ 
For  $i = 1$  to  $n$  do
    Calculate  $Uf_g(t)$  using eqn (10)
    Calculate  $Lf_g(t)$  using eqn (11)
    Initialize  $U_I(t)$  using eqn (12)
    Initialize  $L_I(t)$  using eqn (13)
    Update  $UT_c(t)$  by calculating elementwise product of previous cell state and input gate using eqn (14)
    Update  $LT_c(t)$  by calculating elementwise product of new candidate value using eqn (15)
    Calculate and update lower and upper cell state using eqn (16) and eqn (17)
    Provide weather prediction output for both  $UO_g(t)$  and  $LO_g(t)$  using eqn (18) and eqn (19)
    Calculate hidden state by combining the output of  $UO_g(t)$  and  $LO_g(t)$  using eqn (20)
    Return  $H(t-1)$ 
End for

```

The mathematical representation of the gates are defined as follows,

$$Uf_g(t) = \alpha(wUf_g \cdot [H(t-1), I(t)] + BUf_g) \quad (10)$$

$$Lf_g(t) = \alpha(wLf_g \cdot [H(t-1), I(t)] + BLf_g) \quad (11)$$

$$U_I(t) = \alpha(wU_I \cdot [H(t-1), I(t)] + BU_I(t)) \quad (12)$$

$$L_I(t) = \alpha(wL_I \cdot [H(t-1), I(t)] + BL_I(t)) \quad (13)$$

$$UT_c(t) = \tanh[wUT_c \cdot [H(t-1), I(t)] + BUT_c] \quad (14)$$

$$LT_c(t) = \tanh[wLT_c \cdot [H(t-1), I(t)] + BLT_c] \quad (15)$$

Where, α is a sigmoid function and w represent the weight value of the gates in the C-LSTM. Then the upper $Uf_g(t)$ and lower $Lf_g(t)$ forget gate is assorted by element wise product with the time step of previous cell state and the upper $U_I(t)$ and lower $L_I(t)$ input gate is assorted by element wise product with the step of the new temporary upper $UT_c(t)$ and lower $L_I(t)$ candidate value. The sum of these gates are defined as follows,

$$UC(t) = Uf_g(t) \odot UC(t-1) + U_I(t) \odot UT_c(t) \quad (16)$$

$$LC(t) = Lf_g(t) \odot LC(t-1) + L_I(t) \odot LT_c(t) \quad (17)$$

The final gate is lower $LO_g(t)$ and upper $UO_g(t)$ which provide the information of the output. The mathematical representation of the output gate is defined as follows,

$$UO_g(t) = \alpha(wUO_g \cdot [H(t-1), I(t)] + BUO_g) \quad (18)$$

$$LO_g(t) = \alpha(wLO_g \cdot [H(t-1), I(t)] + BLO_g) \quad (19)$$

Finally, the hidden state $H(t-1)$ is calculated by combining the output of both upper and lower gate that is defined as follows,

$$H(t-1) = UO_g(t) * \tanh(UC(t)) + LO_g(t) * \tanh(LC(t)) \quad (20)$$

Using C-LSTM we have perform weather prediction by considering the input P_{wc} and C_{wd} . Based on the input C-LSTM provides the output such as weather prediction report. The weather characteristics predicted by this network are temperature (t), wind speed (w), wind direction (wd), precipitation (p), humidity (h), pressure (δ) dew point (D) etc. These measurements are considered for the purpose of positioning the UAVs and path planning for effective communication. The predicted weather characteristics are also used for effective channel modeling for the purpose of data collection. Fig. 4 represent the process of C-LSTM based weather prediction. Table 3 represent the weather reports for one day (24 h).

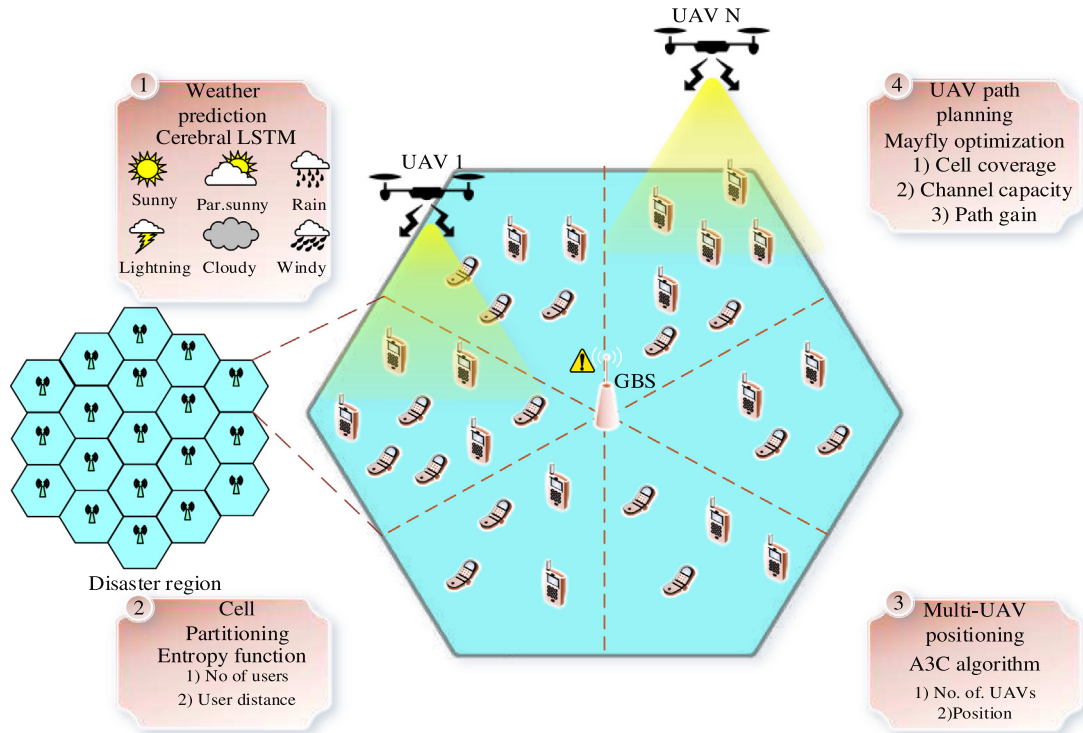


Fig. 3. System model.

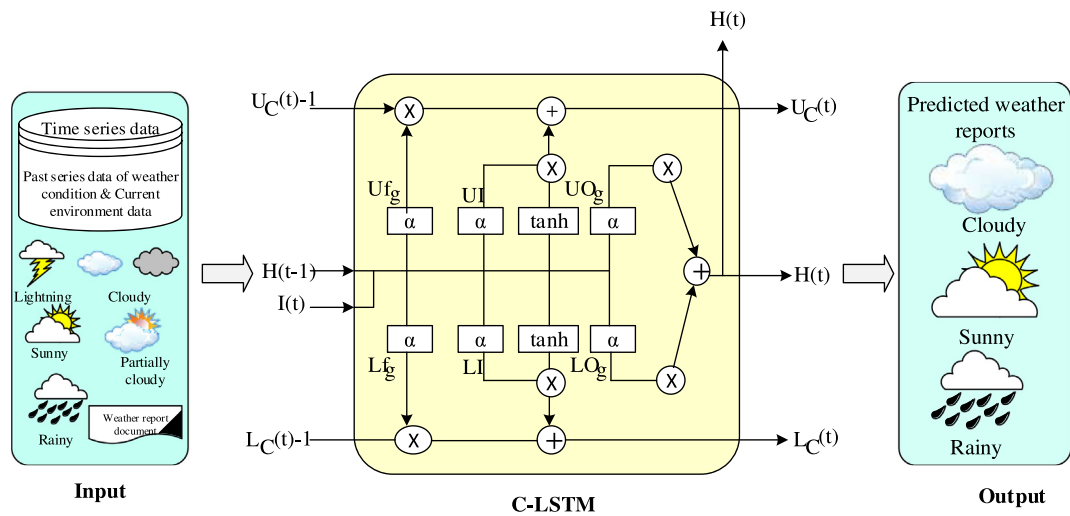


Fig. 4. C-LSTM based weather prediction.

Table 3
Weather prediction report.

T	w	t	p	w & wd	δ	h	D
00:00	Sunny	29 °C	0.1 mm	light wind, 4 m/s from southwest	1010 hPa	74%	23°
03:00	Sunny	26 °C	0 mm	light wind, 4 m/s from southwest	1010 hPa	82%	23°
06:00	Sunny	26 °C	0 mm	light wind, 4 m/s from southwest	1008 hPa	83%	23°
09:00	Cloudy	24 °C	0 mm	light wind, 4 m/s from southwest	1010 hPa	90%	22°
12:00	Cloudy	27°C	0.1 mm	light wind, 5 m/s from west-southwest	1012 hPa	84%	23°
15:00	Cloudy	28 °C	2.0 mm	light wind, 5 m/s from west-southwest	1010 hPa	67%	22°
18:00	Rainy	29°C	0.6 mm	light wind, 4 m/s from southwest	1007 hPa	64%	23°
21:00	Rainy	30 °C	0.7 mm	light wind, 4 m/s from southwest	1007 hPa	67%	22°

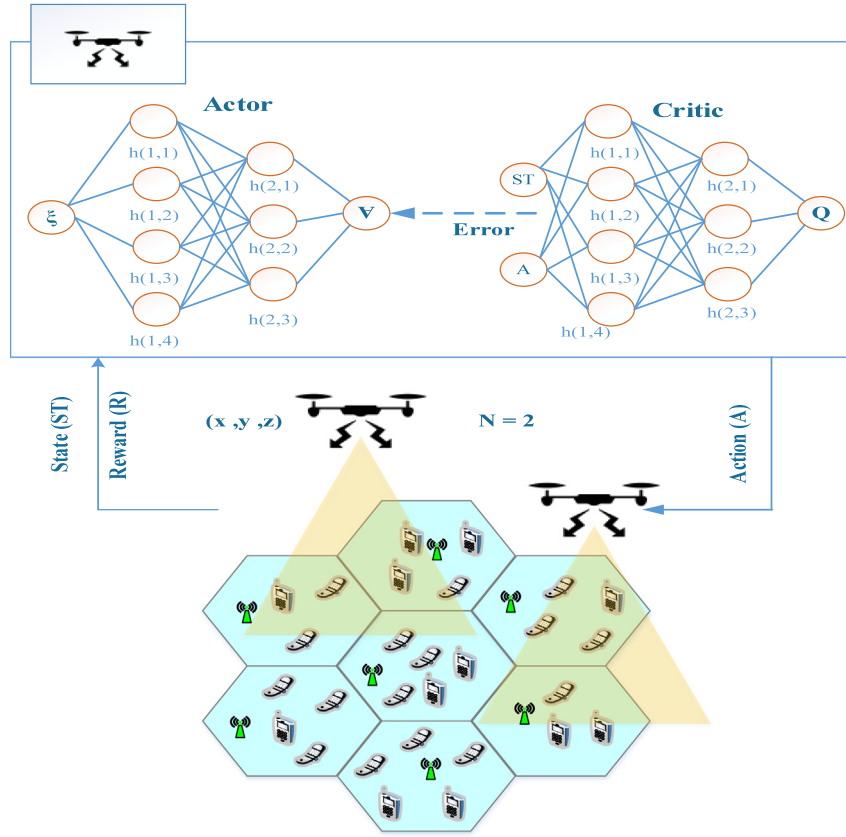


Fig. 6. A3C based Multi-UAV positioning.

The pseudo code for the A3C algorithm based determination of number and position of UAVs in each cell is presented below. The Table 4 presents the state and action of the proposed A3C algorithm. Fig. 6 represents the process of A3C based multi-UAV positioning.

Pseudo code: A3C algorithm

$\pi\theta$ – actor network, C_ϕ^q – critic network, q –

critic scaling loss, γ – batch size, SGD optimizer

Randomly initialize ϕ, θ ;

Use $\pi(\theta)$ to get roll-out size of γ ;

Compute advantage estimation for each transition (ζ) using eqn (29)

Formulation of target by utilizing eqn (31)

Compute critic loss:

$$L_s = \frac{1}{\gamma} \sum_{\zeta} (y(\zeta) - C_\phi^q)^2 \quad (32)$$

Compute critic gradient:

$$\Delta^{critic} = \frac{\partial L_s}{\partial \phi} \quad (33)$$

Compute actor gradient:

$$\Delta^{actor} = \frac{1}{\gamma} \sum_{\zeta} \nabla_{\theta} \log \pi_{\theta}(A|ST) A D^{\pi}(\zeta) \quad (34)$$

Gradient descent calculation using $\Delta^{actor} + q\Delta^{critic}$

Table 4

Entity description.

Entity	Description
State attributes	Density of users (d_u) User pattern ($\mathcal{U}$) Delay sensitivity (d) Predicted weather (p_w) Coverage area (A_c) Interference sensitivity (d) Data rate (r) Elevation angle (θ) Path loss (p_l) LOS/NLOS characteristics (μ)
Action	$a(1)$ = Number of UAVs N_i $a(2)$ = positioning of UAVs (x_i, y_i, z_i)

the cell coverage, channel capacity and path gain. The path planning is performed based on the parameters such as wind characteristics (w), temperature (T), energy (e), speed (s), distance (D), obstacles (o), LOS/NLOS (μ), optimality (τ), completeness (c), cost efficiency (c_e) and collision avoidance (α). The proposed MOA includes the feature combination of GA, PSO and firefly optimization algorithm. The position of every mayfly represents the possibility of obtaining an optimal solution. Here, UAV is represented as mayfly and it selects the optimal path for reducing the energy consumption. Group of male and female mayflies are generated randomly, hence the initial position of the mayflies are randomly selected in the search space that is defined as follows,

$$X = [x_1, x_2, \dots, x_n] \quad (35)$$

Where X represent the position of the male mayfly (mf). The position vector performance is measured by the objective function. The velocity of mayflies are defined as follows,

$$V = [v_1, v_2, \dots, v_n] \quad (36)$$

5.4. Energy efficient path planning

The UAV path planning is an important process in effective communication in which the path of the UAV during movement is planned optimally in order to reduce the energy consumption of the UAV. The path planning of UAV is affected by various factors such as weather condition, limited energy availability and presence of blockages. The MOA is used to optimally plan the path of the UAV in order to maximize

Where, V represent the velocity which is used to update the position of the mayfly. The search agent update their position depend upon their own appropriate position that is known as P_{best} , the appropriate position is updated by other mayflies are known as g_{best} . The collection of male mayflies in a swarm demonstrates that their positions are updated based on the individual and social experiences. The position of the male mayfly is defined as follows,

$$x_i(t+1) - v_i(t+1) + x_i(t) \quad (37)$$

Where, $x_i(t)$ represent the present position of the i th mayfly. $x_i(t+1)$ and $v_i(t+1)$ represent the position and velocity of the i th mayfly in next stage. The dance of mayfly is performed in height of some meters, and the speed of mayfly is evaluated as follows,

$$v_i d(t+1) = v_{id}(t) + a1 \times \exp(-\epsilon d_p^2) \times (P_{bestdd} - x_{id}(t)) + a2 \times \exp(-\epsilon d_g^2) \times (g_{bestd} - x_{id}(t)) \quad (38)$$

Where, $a1$ and $a2$ are represent the positive attraction constants, and ϵ represent the limits of mayfly visibility to one another. d_p Represent the distance between P_{best} and g_{best} , and v_{id} , x_{id} represent the speed and position with d dimension, where the value of $d = 1, 2, \dots, D$ that is defined as follows,

$$P_{best} = \begin{cases} p_i(t+1), & F(p_i(t+1)) < F(P_{besti}) \\ P_{besti}, & F(p_i(t+1)) \geq F(P_{besti}) \end{cases} \quad (39)$$

Where, $f(\cdot)$ represent the objective function that is determined the quality of the solutions. The calculation of d_p^2 and d_g^2 are defined as follows,

$$d_p^2 = \left(\sum_{j=1}^{dmax} (p_{id} - p_{besti}) \right)^{0.5} \quad (40)$$

$$d_g^2 = \left(\sum_{d=1}^{dmax} (p_{id} - g_{best}) \right)^{0.5} \quad (41)$$

$$F = \sum_{i=1}^{dmax} |w, T, e, s, D, \mu| \quad (42)$$

Where, F represents the fitness value that is calculated by considering w, T, e, s, D, μ , in which the optimal path is discovered by the UAV it reaches τ, c, c_e, α . To reserve the optimal path planning the group of appropriate mayflies are continue the nuptial dance with the movement of vertical. Hence, the optimal mayfly keeps its speed change that adds a random feature as defined as follows,

$$v_i(t+1) = v_{id}(t) + nd \times \delta \quad (43)$$

Where, nd represent the nuptial dance and δ represent the random value from -1 to 1 . Then the female mayflies are flying towards the male mayflies for performing the process of mating. The population initialization of female mayfly (ff) is defined as follows,

$$Y = [y1, y2, \dots, ym] \quad (44)$$

Where Y represent the position of ff . And the velocity of ff is defined as follows,

$$U = [u1, u2, \dots, un] \quad (45)$$

The updated position of female mayfly is defined as follows,

$$y_i(t+1) - v_i(t+1) + y_i(t) \quad (46)$$

The process of attraction is performed in which the appropriate female is attracted to the appropriate male. This process is continued till end. The velocity calculation of female mayfly is defined as follows,

$$v_i d(t+1) = \begin{cases} v_{id}(t) + a2 \times \exp(-\epsilon d_f^2) \times (x_{id}(t) - y_{id}(t)), & F(y_i) > F(x_i) \\ u_{id}(t) + a_n \times \delta, & F(y_i) \leq F(x_i) \end{cases} \quad (47)$$

Where, y_{id} and u_{id} represent the female mayfly position and speed in d th dimension respectively. d_f^2 is represent the distance between male and female mayflies and a_n represents the random coefficient. The optimal matching process of female and male is based on the objective function and fitness values. The crossover offspring is calculated as follows,

$$\alpha_1 = \sigma \times mf + (1 - \sigma) \times ff \quad (48)$$

$$\alpha_2 = \sigma \times ff + (1 - \sigma) \times mf \quad (49)$$

Where, α_1 and α_2 are represent the offspring 1 and 2, σ represent the random number within a range. mf represent the male mayfly and ff represent the female male fly. By doing so the robust path planning is achieved thereby increasing the energy efficiency of the system. Fig. 7 represents the flowchart of the mayfly optimization.

Pseudocode for Mayfly Optimization

```

Initialize the population of male mayfly  $X = \{x1, x2, \dots, xn\}$  with velocity
 $V = \{v1, v2, \dots, vn\}$ 
Initialize the population of female mayfly  $Y = \{y1, y2, \dots, ym\}$  with
velocity  $U = \{u1, u2, \dots, un\}$ 
Estimate the solutions by predefined objective functions
SORT the mayflies
Do While stopping criteria are not met
    Update position and velocities of  $mf$  using eqn (35) and eqn (36)
    Update position and velocities of  $ff$  using eqn (44) and eqn (45)
    If current mayfly dominated its  $P_{best}$  using eqn (39)
        Replace the  $P_{best}$  with the current solution
    Else no one dominated others
        The current solutions is replace the  $P_{best}$ 
    End if
    MATE the mayflies
    Calculate offspring using eqn (48) and eqn (49)
    Separate offspring for  $mf$  and  $ff$  randomly
    If an offspring dominates the same sex parent
        Replace parent with the current offspring
    End if
    Insert all the current non-dominated solutions to detect the external
    repository
    SORT the non-dominated solutions and prune the repository if need
End while
Return current offspring for discover optimal path planning

```

6. Experimental study

The experimentation of the proposed IWPOP-UAV is carried out in this section. This section is further divided into three sub-sections namely simulation setup, comparative analysis, and research summary. The superiority of the proposed approach is proved by executing an extensive analysis.

6.1. Simulation setup

The proposed approach is simulated by utilizing the Network Simulator (NS 3.26) simulation tool. The system configuration required for the simulation of this approach is provided in Table 5. Initially, the prediction of weather is carried out based on the historical weather report and based on that the positioning and path planning of UAV is carried out in order to maximize the coverage, and minimize flight time and energy consumption. The parameters required for the implementation of our approach is presented in Table 6.

6.2. Usecase

The Proposed IWPOP-UAV model is demonstrated with a use case as depicted in Fig. 8. The environment considered is affected by fire along with the GBS. This hinders the communication of UEs in that region in reporting emergency situation. In this situation, the number of UAV-BS required to provide uninterrupted communication to the UEs

Table 5

System configuration.

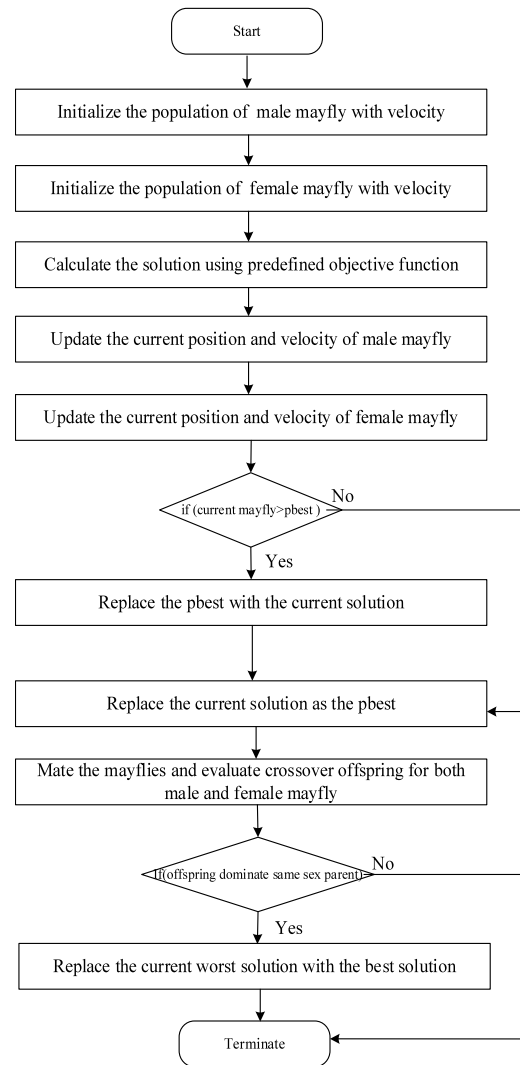
Hardware specifications	Processor Hard Disk RAM	Pentium Dual Core and Above 60 GB 8 GB
Software Specifications	OS Network Simulator	Ubuntu 14.04 LTS NS3.26

Table 6

Simulation parameters.

Parameters	Description
Network Parameters	
Simulation area	2000 m*1000 m*880 m
Number of GBS	1
Number of UAVs	10
Number of UEs	100
Wind speed	6 m/s
Buffer distance	3 m
Path loss constant	3
Bandwidth	1 MHz
Noise power	−170 dBm
Transmit power of UEs	12 mW
Transmit power of UAVs	4.46 W
Wavelength	1.45 m
Data bit rate	300 kbps
Carrier frequency constant	12
Packet size	114 Bytes
Path width	12 m
Number of blades	4
Blade chord	0.2 m
Air density	1.234 kg/m ³
Response time constant	0.987
Flying time	3 min
Path gain	−13.3 dB
Cell coverage	0.93
Capacity	70.32 kbps
QoS	0.92
Algorithm Parameters	
C-LSTM	
Epoch	450
Learning rate	0.001
Optimization	Adagrad
Batch size	128
State size	512
A3C	
Actor learning rate	0.001
Critic learning rate	0.001
entropy	0.01
Discount factor	0.95
Hidden layers	2
Mayfly Optimization	
η_1	1
η_2	1.5
γ	2
nd	0.1
f_r	0.1

is determined and the position of the UAVs in the emergency region is determined based on the predicted weather conditions. The ground monitoring center is responsible for the placement and controlling of

**Fig. 7.** Flowchart for Mayfly optimization.

UAVs. The consideration of weather conditions facilitates the optimal positioning and path planning of UAVs. The UAVs achieve increased coverage ratio which enables the reliable collection of data without any delay from the UEs. The ground monitoring center receives the collected data from the UAVs and facilitates the reporting of emergency situation. By doing so, the QoS, reliability, and energy efficiency in UAV communication is achieved. Fig. 8 represents the use case diagram.

6.3. Comparative analysis

In this subsection, the proposed approach is validated by comparing it with the existing approaches such as SWIM [39], and multi-rotor UAV [43] by means of several performance metrics. The comparative analysis comprises of three elements namely QoS analysis, reliability analysis, and energy analysis.

(1) QoS analysis

The QoS of an approach is a significant metric to be considered in order to determine its efficiency. The QoS of the UAV positioning and path planning can be determined by three different performance metrics namely Coverage ratio, cell coverage and delay. The coverage ratio is referred to as the ratio of covered UEs by the UAV to the total number of UEs present in the emergency situation. The coverage ratio for an approach should be high in order to achieve increased QoS value. The analysis of coverage ratio of the proposed IWPOP-UAV approach

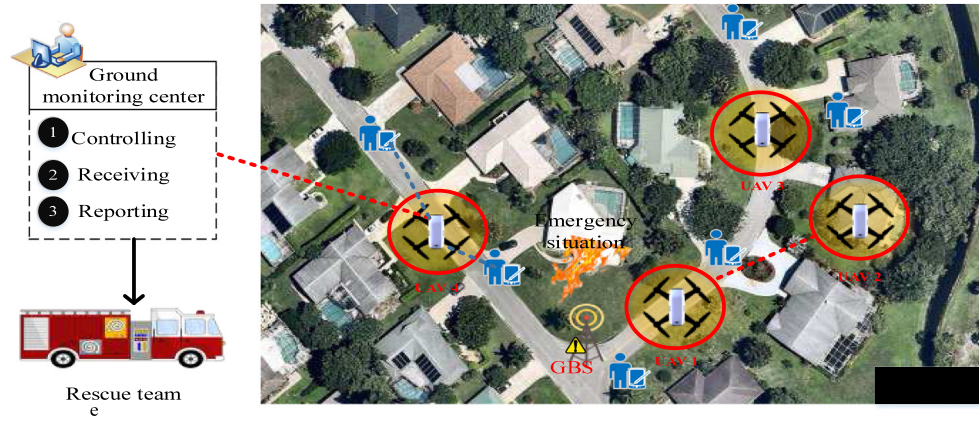


Fig. 8. Use-case diagram for emergency scenario.

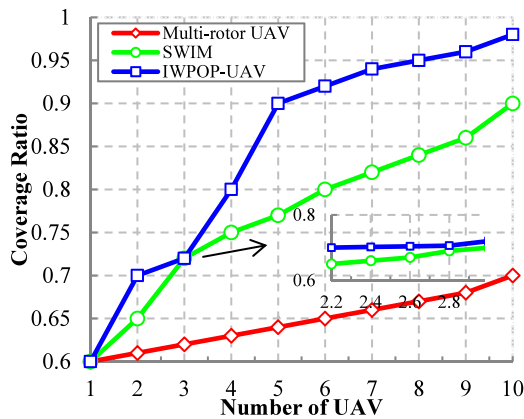


Fig. 9. Coverage ratio vs. number of UAVs.

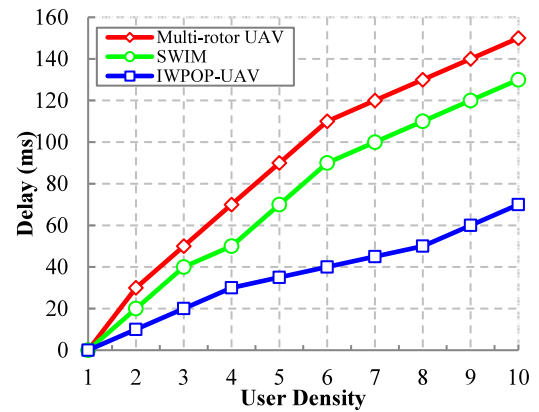


Fig. 11. Delay vs. user density.

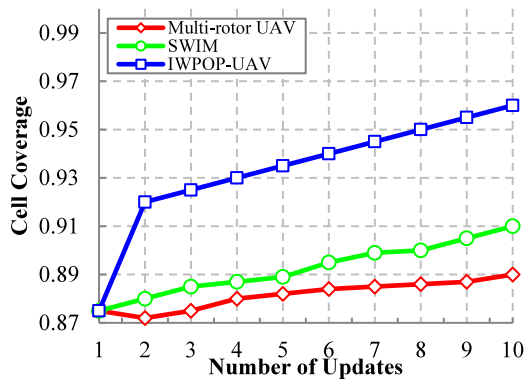


Fig. 10. Cell coverage vs. number of updates.

and the existing approaches with respect to number of UAVs is depicted in Fig. 9. From the figure it can be said that the coverage ratio is directly proportional to the quantity of UAVs. The coverage ratio of the proposed approach is high due to the determination of number of UAVs required along with its positioning. The A3C algorithm is utilized for this purpose which functions based on the number of UEs in the cell.

The existing approaches lack in determining the number of UAVs required for effective coverage of UEs. The cell coverage is referred to as the coverage possessed by the UAV during its motion in collection of data from the UEs in emergency situations. The analysis of cell coverage of the proposed approach and the existing approaches with respect to number of updates is illustrated in Fig. 10. The cell coverage increases linearly with increase in number of updates of the position.

The proposed approach possesses increased cell coverage than the existing approaches due to the optimal planning of UAV path in each and every position updates in order to cover large number of UEs.

The proposed mayfly optimization considers the weather conditions which are predicted in the previous process along with the available energy and presence of blockages in the environment thereby resulting in realistic coverage of UEs in the cells. The existing approaches lack in consideration of weather conditions which greatly influences the trajectory of the UAV resulting in hardships to the UAV. Further the presence of obstacles in the environment is not considered by the existing approaches resulting in attenuations caused by NLOS conditions. The delay in collection of data from the UEs affects the QoS greatly. Particularly in emergency situations the delay should be very much low. The analysis of delay produced by the proposed approach and the existing approaches with respect to user density is presented in Fig. 11. The delay increases with increase in the density of users. The proposed approach possesses low delay than other approaches due to the dynamic determination of number of UAVs based on the increasing number of UEs in the cells. The positioning and path planning of UAV is also optimized which provides timely service in emergency situations. The existing approaches deployed random number of UAVs with considering the user density which resulted in increased delay when the scalability of UEs is increased,

The numerical analysis of QoS of the proposed IWPOP-UAV approach and the existing approaches is carried out in terms of several QoS metrics such as coverage ratio, cell coverage, and delay which is shown in Table 6. The proposed approach possesses 0.847, 0.934, and 36.5 ms of coverage ratio, cell coverage and delays respectively whereas the existing approaches 0.64 to 0.71, 0.82 to 0.89, and 73.5

Table 7
Analysis of QoS.

Methods	Coverage ratio	Cell coverage	Delay (ms)
	Number of Updates	Number of UAV	User density
Multi-rotor UAV	0.646 ± 0.5	0.882 ± 0.3	89 ± 0.4
SWIM	0.771 ± 0.4	0.893 ± 0.2	73.5 ± 0.3
IWPOP-UAV	0.847 ± 0.2	0.934 ± 0.1	36.5 ± 0.1

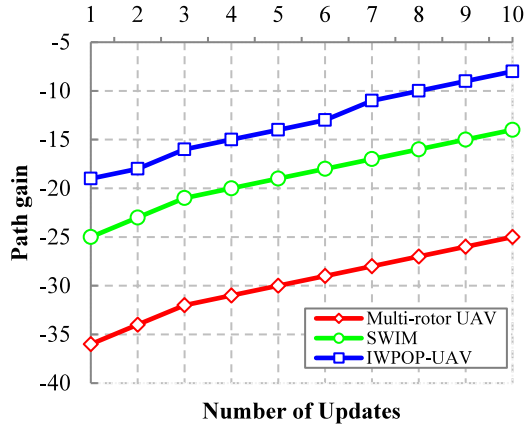


Fig. 12. Path gain vs. number of updates.

to 89 ms of coverage ratio, cell coverage and delay respectively. From this we can conclude that the proposed approach possesses increased coverage ratio and cell coverage, and minimal delay than the existing approaches due to joint optimization of position and path planning of UAV hence resulting in increased QoS in UAV communication.

(2) Reliability Analysis

The reliability of an approach is a countable metric to be considered in order to determine the efficacy of it. The reliability is computed by analyzing the performance metrics such as path gain and number of collected packets from the UEs in the environment. The path gain is referred as the positive metric which should be large in order to achieve the reliable collection of data. The path gain of the proposed approach and the existing approaches is analyzed with respect to number of updates as depicted in Fig. 12. The path gain increases with increase in the number of updates. The proposed approach seems to possess increased path gain than other approaches due to the implementation of optimal path planning of UAV. The optimal path of the UAV is influenced by presence of obstacles, adverse weather conditions, and energy availability of UAVs for performing collection of data and motion. These multiples parameters are considered by the proposed approach in order to achieve the optimal path, the existing approaches does not consider all these parameters which affects the path gain.

The number of collected packets by the UAVs denotes the successful transmission of packets from UEs to the UAV contributing to increased reliability. The collection of packets by the UAV is affected by two factors namely speed and altitude of the UAV. Fig. 13 depicts the analysis of quantity of collected packets of the proposed approach and the existing approaches with respect to varying speed of UAVs. From the figure it can be said that the increase in speed of the UAV affects the collection of packets. The proposed approach possesses increased number of collected packets due to the increased coverage ratio achieved by it. The path planning of UAVs by implementing Mayfly optimization based on the multiple parameters such as blockage, energy and weather conditions supports increased collection of packets whereas the existing approaches achieves reduced number of collected packets due to the NLOS conditions and adverse weather conditions.

The analysis of number of collected packets of the proposed approach and existing approaches with respect to varying altitude of UAV is illustrated in Fig. 14. The increased number of collected packets by

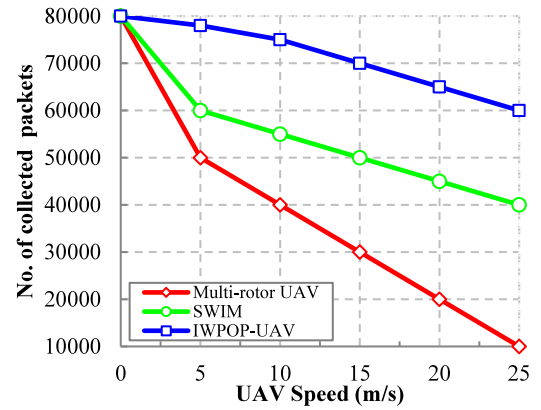


Fig. 13. Number of collected packets vs. UAV speed.

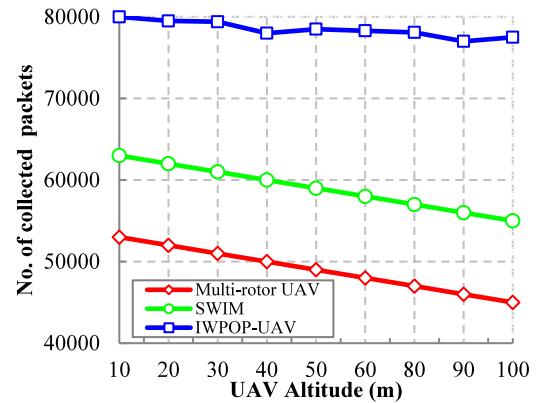


Fig. 14. Number of collected packets vs. UAV altitude.

the proposed approach is high due to the optimal positioning of UAVs in the network. This is carried out by using A3C algorithm in which the weather conditions predicted by the C-LSTM are considered along with several significant parameters such as sensitivity to interference and delay. The parameters are utilized to determine the optimal altitude of the UAV resulting in increased number of collected packets. The existing approaches performed ineffective position of UAVs which is hindered by increased interference and path loss affecting the collection of packets.

The Table 7 presents the numerical analysis of the reliability of the proposed IWPOP-UAV approach and the existing approaches. The analysis is executed in terms of reliability factors such as path gain and number of collected packets. The proposed approach possesses about -13.3 dB of path gain and 71,333 to 78,477 of average number of collected packets whereas the existing approaches possess about -18.8 dB to -29.8 dB of path gain and 38,333 to 59,200 number of collected packets. It is seen that the proposed approach achieves 5.5 dB of path gain greater than SWIM and 16.5 dB greater than Multi-rotor UAV. From this we can conclude that the proposed approach is reliable in collection of data from the UEs in emergency situations.

(3) Energy Analysis

The energy of the UAVs is limited and hence it should be efficiently managed by an approach. The Energy efficiency is referred to as the measure of lifetime of the UAV in performing a particular task. The energy analysis is performed by means of two performance metrics namely UAV transmit power and energy consumption. The UAV transmit power is a significant metric to be considered in the multi-UAV scenario due to the interference power between the UAVs in collection of data. The positioning of UAV influence the transmit power of the UAV, if the positioning of UAV is performed in an optimal way then the

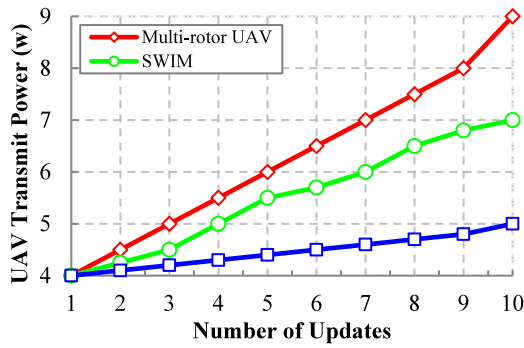


Fig. 15. UAV transmit power vs. number of updates.

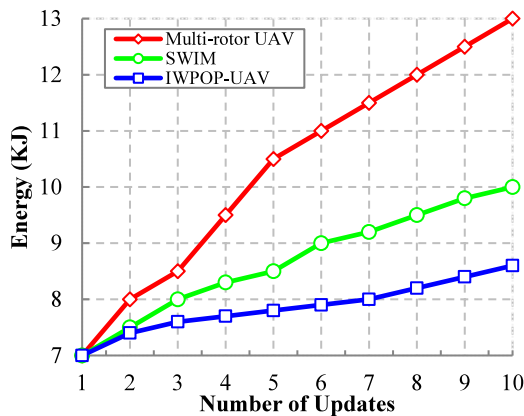


Fig. 16. Energy vs. number of updates.

transmit power required for collection of data is minimized, similarly if the positioning of the UAV is performed inefficiently then the collection of data is affected even if the maximum transmit power is utilized. The analyses of UAV transmit power of our proposed approach and the existing approach with respect to number of updates is presented in Fig. 15. The UAV transmit power of our proposed approach is found to be minimized due to the positioning of the UAVs by implementing A3C algorithm. The consideration of significant parameters and execution of dynamic positioning achieves the stable collection of data with reduced UAV transmit power. The ignorance of important parameters such as weather conditions affects the positioning of UAV in the existing approaches which thereby increases the required UAV transmit power.

The energy consumption of UAVs denotes the total energy consumption in motion and communication of the UAVs in performing collection of data. The path planning of the UAVs affects the consumption of energy which should be optimized to minimize the energy consumption. The energy analysis of the proposed and existing approaches with respect to number of updates is presented in Fig. 16. The energy consumption of the proposed approach is minimal than other approaches due to the implementation of MOA based path planning. The consideration of obstacles and weather conditions in the environment facilitates effective planning of path of the UAVs. The existing approaches lack in performing realistic path planning of UAV thereby resulting in increased energy consumption.

The energy analysis of the proposed approach and existing approaches in a numerical manner is presented in Table 8. The numerical analysis is performed by means of energy metrics such as UAV transmit power and energy consumption of UAVs. The proposed approach possesses about 4.46 W and 7.86 kJ of UAV transmit power and energy consumption respectively whereas the existing approaches possess about 5.52 W to 6.3 W and 8.68 kJ to 10.35 kJ of UAV transmit power

Table 8

Analysis of reliability.

Methods	Path gain	Number of collected packets	
	Number of Updates	UAV speed (ms)	UAV altitude (m)
Multi-rotor UAV	-29.8 ± 0.4	38333.3 ± 0.5	49122.3 ± 0.3
SWIM	-18.8 ± 0.3	55000.4 ± 0.4	59200.5 ± 0.2
IWPOP-UAV	-13.3 ± 0.1	71333.3 ± 0.1	78477.9 ± 0.1

Table 9

Analysis of energy efficiency.

Methods	UAV transmit power (w)	Energy (KJ)
	No. of Updates	No. of Updates
Multi-rotor UAV	6.3 ± 0.05	10.35 ± 0.04
SWIM	5.525 ± 0.03	8.68 ± 0.03
IWPOP-UAV	4.46 ± 0.01	7.86 ± 0.01

Table 10

Analysis of Time complexity.

Process	Time complexity	Description
Weather prediction	$O(K)$	K denotes the weight of parameters
Multi-UAV positioning	$O(N A)$	N denotes number of iterations
Energy efficient path planning	$O(U*V*W*X)$	U, V, W, X denotes the maximum iteration, size of population, maximum search iterations, time involved in computation of fitness.

and energy consumption respectively. The proposed approach possesses average reduction of 1.5 W and 2 kJ than the existing approaches. From this we can conclude that the proposed approach is energy efficient in performing collection of data (see Table 9).

6.4. Research summary

The experimentation and evaluation of the proposed IWPOP-UAV approach is performed in a brief manner from which the summarization of the results is carried out. The major requirements of UAV based communication are QoS, reliability, and energy efficiency which are achieved by performing optimal positioning and path planning of UAVs. The consideration of weather conditions greatly facilitates the accurate positioning and determination of altitude of the UAVs in the environment resulting in accomplishment of QoS and reliability through increased coverage ratio, number of collected packets etc. providing answer to RQ1. The RQ2 regarding lifetime of the UAVs in the network is answered by performing energy efficient path planning in which the realistic environment of emergency situations is considered to achieve minimal UAV transmit power and energy consumption. The proposed approach outperformed the existing approaches with respect to all the metrics considered for analysis. Table 10 represents the time complexity of the proposed approaches.

7. Conclusion and future work

In this paper, the challenges in UAV communication caused by diverse weather conditions is addressed by proposing a weather condition aware positioning and path planning of UAVs. The major objective of this approach is to maximize the coverage and minimize the energy consumption of UAV in order to achieve increased QoS, reliability and energy efficiency. The prediction of weather condition is performed by using C-LSTM based on the historical data in order to facilitate effective communication. The emergency environment is partitioned into cells to facilitate precise identification of target UEs. The determination of number and positioning of the UAVs in each cell is executed by the

A3C algorithm based on the predicted weather condition and other significant factors. By doing so, the coverage ratio is increased which further results in increased number of collected packets and reduces the delay and required transmit power of UAVs. The path panning of the UAVs is considered as a multi-objective optimization problem and is executed by implementing MOA. This optimal path planning increases the cell coverage and path gain, and minimizes the energy consumption of the UAVs. Hence the lifetime of the UAVs is improved. The proposed approach is experimented in NS 3.26 and evaluated in terms of QoS analysis, Reliability analysis, and energy analysis. The results summarizes that the proposed approach increase 0.25 coverage ratio, increase 0.07 cell coverage, reduce 70 ms delay, increase –25 path gain, reduce 4 w UAV transmit power, reduce 5 kJ energy, collects high packets (20000) with respect to UAV speed and collect high packets (30000) with respect to UAV altitude. In future, the proposed approach will be developed to implement energy harvesting in order to improve the lifetime of the UAV.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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